

JAKUB KIERZNOWSKI

Junior Java Backend Developer | Software Developer Intern at IBM

jakub.kierznowski@gmail.com | +48 662 977 139 | Kraków, on-site / hybrid

github.com/qualv13 | linkedin.com/in/jakub-kierznowski | qualv13.github.io | instaldesk.pl

Java and Spring Boot backend developer. At IBM I work on Data Lineage, optimising graph and tree traversal over enterprise metadata. After hours I run InstalDesk, a B2B SaaS I built alone, with a native integration to the Polish KSeF e-invoicing system. Final-year Computer Science and Intelligent Systems at AGH, GPA 4.74/5.00, graduating January 2027.

EXPERIENCE

Software Developer Intern

IBM Software Lab · Kraków · hybrid
02.2026 – present

Backend work in Java and Spring on Data Lineage. I optimise graph and tree traversal over large enterprise metadata graphs. Wrote the automated pytest suite for that pipeline with parametrised cases and session-scoped fixtures; coverage went up and the CI run got shorter. I own the Jenkins job scheduling for it. Daily Scrum, English-speaking cross-functional team.

Solo Developer, InstalDesk

Own product · instaldesk.pl
04.2026 – present

Quoting and VAT-invoicing SaaS for Polish installation contractors. A web PWA and a Windows desktop client share one account. Live, currently v1.9.0.

The KSeF integration is native: API v2 against the FA(3) schema, KSeF-token auth on web, XAdES-BES certificate auth on desktop, filed straight to the Ministry of Finance without a third-party integrator. Python and FastAPI on the backend, PostgreSQL in the cloud, SQLite locally so the desktop client keeps working offline. Per-tenant isolation, KSeF tokens encrypted at rest. Stripe billing over BLIK, card and Przelewy24, a 14-day trial without a card, renewal flows, transactional email, PDF generation, XLSX and CSV import and export. I built all of it and I support it.

IT Team Lead, Web & Workflow Automation

Student Council · WRSS WEAlIB, AGH
03.2024 – present

I maintain the council website (Vue, TailwindCSS) and the VPS it runs on, and built internal automations on self-hosted n8n against Slack and Google APIs over OAuth 2.0. Event coordinator for the VOLT 2025 tech conference, and part of the organising team for VOLT 2026.

Team Mentor → Community Leader

Zwolnieni z Teorii · remote
09.2022 – 05.2025

Mentored 35+ student teams per edition, from planning to the final presentation of their social-impact projects. Later ran a participant community (ZWzT × PwC certificate) and supported olympiad entrants.

Game Developer

AGH Code Industry · 10.2023 – 06.2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game built by a student team under a mentor.

MY PROJECTS

DrugPath, biomedical knowledge-graph agent

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

An agent that walks the Hetionet biomedical graph (47,031 nodes, 293k relationships) along drug → gene → pathway → disease paths to explain how a drug works and where it might be repurposed. Educational project on public research data. Demo at qualv13.github.io/neo4j-agent.

IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Backend for managing smart-lamp fleets. JWT auth with refresh tokens, MQTT command and telemetry topics over RabbitMQ, a Protocol Buffers device protocol, OTA firmware delivery from S3, scheduled device jobs. Layered Spring architecture with 15 JUnit 5 test classes covering controllers, services, repositories and security, integration tests on Testcontainers, and a full Docker Compose stack.

Green-window EV charging optimiser

CarCharging · Java, Spring Boot, REST, Docker

Finds the cleanest 48-hour charging window from the UK Carbon Intensity API with a sliding window over half-hourly forecasts, and serves the forecast through a REST API.

BPM-first workout playlist generator

Fit2Beat · HackYeah 2025, top 8 of 65+ · Python, Flask, Docker

Builds playlists from a target BPM rather than from genre. Nobody else in the biohacking track approached it that way. Team of four, 24 hours, 3.9/5 from the jury.

SKILL SET

Tech

Core. Java 17, Spring Boot, Spring Security, Hibernate/JPA, Maven, JUnit 5, Testcontainers, PostgreSQL, SQL, REST APIs.

Working knowledge. Python (FastAPI, pytest), Docker and Compose, Linux/Bash, Git, Jenkins CI/CD, RabbitMQ, MQTT, Protocol Buffers, OAuth 2.0, Neo4j and Cypher, AWS S3, algorithms and data structures.

Also used. Vue with TailwindCSS, C# (Unity), C++, n8n, LLM APIs, self-hosted models (Ollama), TensorFlow, scikit-learn.

Soft

Agile and Kanban, daily Scrum in a cross-functional English-speaking team. Team mentor at Zwołnieni z Teorii for three editions, 35+ teams each. Lead event coordinator for the VOLT 2025 conference, including partnership deals with tech companies. I ship my own product and handle its support and user contact myself.

Languages: Polish native, English C1 (working language at IBM), French A2.

EDUCATION

Computer Science and Intelligent Systems, BEng

AGH University · Kraków

10.2023 – 01.2027

GPA 4.74/5.00. Year Representative since the 1st semester, Student Council since the 2nd. Engineering thesis: applying deep learning to 2D-image to 3D-mesh conversion.

Extended Mathematics, Physics and Computer Science

V High School · Kraków

2019 – 2023

C++, competitive-programming problem solving.

CERTIFICATIONS

Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

INTERESTS

Snowboarding, sailing, a self-hosted homelab with Docker and quantised local models on a Raspberry Pi and a VPS, game dev, board games.